

Tutorial 3 - Answers

Question 1

Proof.

Pairwise disjoint. Fix $i \neq j$. Then

$$(A \cap X_i) \cap (A \cap X_j) = A \cap (X_i \cap X_j) = A \cap \emptyset = \emptyset$$

Union equals A. Check both inclusions.

($\subset$) For each i , $A \cap X_i \subset A$ so $\bigcup_{1 \leq i \leq n} (A \cap X_i) \subset A$.

($\supset$) Let $x \in A$. Since $A \subset X$, we have $x \in X$, and since $X = \bigcup_{1 \leq i \leq n} X_i$, there exists some k with $x \in X_k$. Then $x \in A$ and $x \in X_k$ so $x \in A \cap X_k \subset \bigcup_{1 \leq i \leq n} (A \cap X_i)$.

Combining both inclusions:

$$\bigcup_{1 \leq i \leq n} (A \cap X_i) = A$$

Since $A \cap X_1, \dots, A \cap X_n$ are pairwise disjoint with union A , they form a partition of A . ■

Question 3

Let 0 denote the zero function, $0(x) = 0$ for all x .

Proof (Claim 1). By definition,

$$[0] = \{g \in A : 0 \text{ R}g\} = \{g \in A : \exists c \in \mathbb{R} \text{ s.t. } 0(x) - g(x) = c, \forall x\}$$

The condition $0(x) - g(x) = c$ for all x says $-g(x) = c$ for all x , that is, $g(x) = -c$ or all x , a fixed constant since c does not depend on x . Conversely, if $g(x) = k$ for all x , (some fixed $k \in \mathbb{R}$), then $0(x) - g(x) = -k$ for every x , so $0 \text{ R}g$ with constant $c = -k$. Hence

$$[0] = \{g \in A : g(x) = k, \forall x, \text{ for some } k \in \mathbb{R}\},$$

that is, $[0]$ is exactly the set of constant functions. ■

Proof (Claim 2). By definition,

$$[f] - \{g \in A : fRg\} = \{g \in A : \exists c \in \mathbb{R} \text{ s.t. } f(x) - g(x) = c, \forall x\}$$

The condition $f(x) - g(x) = c$ for all x is equivalent to $g(x) = f(x) - c$ for all x , that is, $g = f + k$ where $k = -c$ ranges over all of $\mathbb{R}$ as c does. Hence

$$[f] = \{f + k : k \in \mathbb{R}\}$$

■

Question 4

Setup. Recall from Exercise 4.18 that $A = \mathbb{R}^2$ and the original relation is $(x, y) R (u, v) \iff |x| + |y| = |u| + |v|$, whose equivalence classes are “diamonds” centered at the origin. Now consider the new relation R' on A defined by

$$(x, y) R' (u, v) \iff \sqrt{x^2 + y^2} = \sqrt{u^2 + v^2}.$$

Proof (Claim 1). Let $(x, y), (u, v), (a, b) \in \mathbb{R}^2$.

Reflexive? $\sqrt{x^2 + y^2} = \sqrt{x^2 + y^2}$ trivially so $(x, y)R'(x, y)$.

Symmetric? If $(x, y)R'(u, v)$ then $\sqrt{x^2 + y^2} = \sqrt{u^2 + v^2}$ and this is symmetric in the two sides so $(u, v)R'(x, y)$.

Transitive? If $(x, y)R'(x, y)$ and $(u, v)R'(a, b)$ then $\sqrt{x^2 + y^2} = \sqrt{u^2 + v^2}$ and $\sqrt{u^2 + v^2} = \sqrt{a^2 + b^2}$ so by transitivity, $(x, y)R'(a, b)$.

Hence R' is an equivalence relation.

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Proof (Claim 2). By definition,

$$[(x_0, y_0)] = \{(u, v) \in \mathbb{R}^2 : \sqrt{u^2 + v^2} = \sqrt{x_0^2 + y_0^2} = r\}$$

which is exactly a circle of radius r about the origin. If $r = 0$ then

$$\{(u, v) : u^2 + v^2 = (0, 0)\}$$

which is a single point.

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